Problem Set: Problem 2(ii)

Converting $e^-\mu^- \to e^-\mu^-$ to $q\,q' \to q\,q'$ via Gluon Exchange

PHYS 7502 — Particles and Nuclei 2

1 Question Statement and Deliverables

Problem 2(ii) — Verbatim

Also convert the Mandelstam variable amplitude to the equivalent one for $qq \to qq$ parton–parton (parton–parton x.s., ignoring/not including pdf and fragmentation effects) scattering via gluon exchange—presume different quark flavors to avoid the u-channel inclusion. In this case you may either sum/average, or choose specific color choices for the incoming and outgoing quarks and exchanged gluon.

What This Problem Wants — In Plain English

In part (i) you worked out $\langle|\mathcal{M}|^2\rangle$ for $e^-\mu^- \to e^-\mu^-$ (Griffiths 2nd ed. Problem 7.38) and rewrote it in Mandelstam variables s, t, u. Now:

- **Take that Mandelstam-variable result** and convert it to the QCD process $q\,q' \to q\,q'$ (two quarks of *different* flavor scattering via single gluon exchange).
- **Handle the color algebra**: either (a) sum over final colors and average over initial colors to get a color-averaged $\langle|C|^2\rangle$, or (b) pick specific colors (e.g. red quark, green quark, specific gluon) and compute the color factor for that channel.
- **Ignore** parton distribution functions and fragmentation — treat the quarks as free, on-shell, massless particles.
- **Deliverable**: The spin- and color-averaged $\langle|\mathcal{M}|^2\rangle_{qq'\to qq'}$ in Mandelstam variables.

2 Starting Point: The QED Result from Part (i)

Recap: $e^-\mu^- \to e^-\mu^-$ in Mandelstam Variables

From part (i) (Griffiths 2nd ed., Problem 7.38(a), starting from eq. 7.126), we derived the spin-averaged amplitude squared for electron–muon scattering. There is only one diagram — the t-channel photon exchange — and the result in the CM frame at high energy (m, $M \to 0$) is (Griffiths 2nd ed., answer to 7.38a):

$$\langle|\mathcal{M}|^2\rangle_{\text{CM}} = 2g_e^4\,\frac{1+\cos^4(\theta/2)}{\sin^4(\theta/2)} \tag{1}$$

In Mandelstam variables, using the massless-limit identifications (Griffiths 2nd ed., Sec-

tion 3.4):

$$s = 4E^2, \qquad t = -2E^2(1-\cos\theta) = -4E^2\sin^2\tfrac{\theta}{2}, \qquad u = -2E^2(1+\cos\theta) = -4E^2\cos^2\tfrac{\theta}{2} \tag{2}$$

we can verify:

$$\frac{1+\cos^4(\theta/2)}{\sin^4(\theta/2)} = \frac{s^2+u^2}{t^2}$$

(since $s^2 = 16E^4$, $u^2 = 16E^4\cos^4(\theta/2)$, $t^2 = 16E^4\sin^4(\theta/2)$). Thus:

$$\boxed{\langle|\mathcal{M}|^2\rangle_{\text{QED}} = 2g_e^4\,\frac{s^2+u^2}{t^2}} \tag{3}$$

where $g_e^2 = 4\pi\alpha$ and $s+t+u=0$ in the massless limit.

Why Only a t-Channel Diagram?

For $e^-\mu^- \to e^-\mu^-$, lepton flavor is conserved at every vertex: an electron line cannot become a muon line. The only topology that connects the two distinct fermion lines is the t-channel exchange. No s-channel (would require $e^-\mu^-$ annihilating) or u-channel (would require $e^- \leftrightarrow \mu^-$ interchange, but they are distinguishable).
The **exact same logic** holds for $q\,q' \to q\,q'$ with different flavors in QCD: gluons are flavor-blind, so a u-quark cannot become a d-quark at a vertex. Only the t-channel gluon exchange survives. (This is precisely why the problem says "different flavors to avoid the u-channel.")

3 Step 1: QED → QCD Feynman Rule Substitutions

The QED amplitude for $e^-\mu^- \to e^-\mu^-$ (Griffiths 2nd ed., eq. 7.106) is:

$$\mathcal{M}_{\text{QED}} = -\frac{g_e^2}{(p_1-p_3)^2}\left[\bar{u}(3)\gamma^\mu u(1)\right]\left[\bar{u}(4)\gamma_\mu u(2)\right] \tag{4}$$

We now convert each ingredient to QCD. The QCD analog (Griffiths 2nd ed., eq. 8.56) for $q_i\,q'_k \to q_j\,q'_l$ is:

$$\mathcal{M}_{\text{QCD}} = -\frac{g_s^2}{4}\,\frac{1}{(p_1-p_3)^2}\left[\bar{u}(3)\gamma^\mu u(1)\right]\left[\bar{u}(4)\gamma_\mu u(2)\right]\sum_{a=1}^{8}\lambda^a_{ji}\,\lambda^a_{lk} \tag{5}$$

where the subscripts i, j label the color of quark line 1 (incoming, outgoing) and k, l label the color of quark line 2.

Where Every Piece Comes From

Let's map each QED element to its QCD counterpart:

Element	QED	QCD	Why
Vertex factor	$ig_e\gamma^\mu$	$i\frac{g_s}{2}\lambda^a_{ji}\gamma^\mu$	SU(3) generator at each vertex
Propagator	$\frac{-ig_{\mu\nu}}{q^2}$	$\frac{-ig_{\mu\nu}\,\delta^{ab}}{q^2}$	Same Lorentz structure; δ^{ab} ensures gluon color matches
2*Coupling	$2{*}g_e^2$	$2{*}\frac{g_s^2}{4}\cdot\lambda^a_{ji}\lambda^a_{lk}$	$g_s/2$ from each vertex $\times$ color matrices
Spinor structure	$[\bar{u}\gamma^\mu u][\bar{u}\gamma_\mu u]$	identical	Color doesn't touch Dirac space

Key point: The δ^{ab} from the gluon propagator contracts the color index a from vertex 1 with the color index b from vertex 2, collapsing the double sum $\sum_{a,b}$ into a single sum $\sum_a$. This is how we get the factor $\sum_a \lambda^a_{ji}\lambda^a_{lk}$ in the amplitude.

Convention Check: Why $g_s/2$ and not g_s?

In Griffiths' convention (2nd ed., eq. 8.38), the quark–gluon vertex factor is

$$\text{vertex} = i\frac{g_s}{2}\lambda^a\gamma^\mu$$

The $\frac{1}{2}$ is there because the SU(3) generators are $T^a = \frac{\lambda^a}{2}$, and Griffiths writes the vertex in terms of the Gell-Mann matrices λ^a rather than the generators T^a. Many other texts (Peskin & Schroeder, Thomson) define the vertex as $ig_sT^a\gamma^\mu = i\frac{g_s}{2}\lambda^a\gamma^\mu$, so the physics is the same — just be consistent.

4 Step 2: Why Color Factorizes from the Spin Traces

The Factorization Theorem

When we square the amplitude and perform Casimir's trick (sum/average over spins), we encounter:

$$\langle|\mathcal{M}|^2\rangle = \frac{1}{4}\cdot\frac{1}{N_c^2}\,\frac{g_s^4}{16}\,\frac{1}{t^2}\underbrace{\sum_{\text{spins}}\left|[\bar{u}(3)\gamma^\mu u(1)][\bar{u}(4)\gamma_\mu u(2)]\right|^2}_{\text{Dirac traces — same as QED}}\times\underbrace{\sum_{\text{colors}}\left|\textstyle\sum_a\lambda^a_{ji}\lambda^a_{lk}\right|^2}_{\text{color factor}}$$

The factorization is exact because:

- The λ^a_{ji} are **numbers** (elements of 3×3 matrices acting in color space).

- The γ^μ are 4×4 matrices acting in **spinor space**.
- These two spaces are completely independent — there is no "color-spin coupling" at tree level.

So the Dirac trace calculation proceeds *identically* to QED and gives us the same $s^2 + u^2$ structure. We then just need to compute the color factor separately and multiply.

Therefore:

$$\langle|\mathcal{M}|^2\rangle_{\text{QCD}} = \underbrace{\frac{g_s^4}{16}}_{\substack{\text{couplings}}} \times \underbrace{\langle|C|^2\rangle}_{\text{color}} \times \underbrace{\frac{8(s^2+u^2)}{t^2}}_{\substack{\text{spin-summed}\\ \text{(not yet averaged)}\\ \text{Dirac traces}}} \tag{6}$$

where the $\frac{1}{4}$ for spin averaging and $\frac{1}{N_c^2}$ for color averaging are absorbed into $\langle|C|^2\rangle$.

Let's carefully unpack each factor.

5 Step 3: The Dirac Traces (Inherited from QED)

From the QED calculation (Griffiths 2nd ed., eqs. 7.126–7.129, with $m = M = 0$):

$$\sum_{\text{spins}} |\mathcal{M}_{\text{QED}}|^2 = \frac{g_e^4}{t^2} \,\text{Tr}[\gamma^\mu \not{p}_1 \gamma^\nu \not{p}_3] \,\text{Tr}[\gamma_\mu \not{p}_2 \gamma_\nu \not{p}_4] \tag{7}$$

Using trace theorems (Griffiths Rules 10, 13):

$$\text{Tr}[\gamma^\mu \not{p}_1 \gamma^\nu \not{p}_3] = 4\left[p_1^\mu p_3^\nu + p_3^\mu p_1^\nu - g^{\mu\nu}(p_1 \cdot p_3)\right] \tag{8}$$

Contracting the two traces and using massless Mandelstam relations:

$$\begin{aligned} p_1 \cdot p_2 &= \tfrac{s}{2}, \quad p_1 \cdot p_3 = -\tfrac{t}{2}, \quad p_1 \cdot p_4 = -\tfrac{u}{2} \\ p_2 \cdot p_3 &= -\tfrac{u}{2}, \quad p_2 \cdot p_4 = -\tfrac{t}{2}, \quad p_3 \cdot p_4 = \tfrac{s}{2} \end{aligned} \tag{9}$$

the result of the contraction is:

$$\sum_{\text{spins}} |\mathcal{M}_{\text{QED}}|^2 = \frac{g_e^4}{t^2} \cdot 32 \cdot \frac{s^2+u^2}{4} = \frac{8\, g_e^4 (s^2+u^2)}{t^2} \tag{10}$$

Spin-averaging (factor of $\frac{1}{4}$):

$$\langle|\mathcal{M}|^2\rangle_{\text{QED}} = \frac{1}{4} \cdot \frac{8\, g_e^4 (s^2+u^2)}{t^2} = \frac{2\, g_e^4 (s^2+u^2)}{t^2} \quad \checkmark \tag{11}$$

The Factor of 32

Where does the "32" come from? Each trace gives a factor of 4 from the trace theorem (Rule 13: $\text{Tr}[\gamma^\mu \not{a} \gamma^\nu \not{b}] = 4[a^\mu b^\nu + b^\mu a^\nu - g^{\mu\nu}(a \cdot b)]$). So $4 \times 4 = 16$ from two traces. Then when you contract the indices μ, ν between the two trace results, the algebra gives a factor of 2, yielding $16 \times 2 = 32$.

The **same** spin-summed Dirac trace factor $\frac{8(s^2+u^2)}{t^2}$ appears in the QCD calculation, because the λ^a matrices don't touch the γ^μ traces.

6 Step 4: Computing the Color Factor

6.1 Option A: Color-Averaged (Sum/Average)

We want:

$$\langle |C|^2 \rangle = \frac{1}{N_c^2} \sum_{\substack{i,j=1 \\ k,l=1}}^{3} \sum_{a,b=1}^{8} \frac{1}{16} \, \lambda^a_{ji} \, \lambda^a_{lk} \, (\lambda^b_{ji})^* (\lambda^b_{lk})^* \tag{12}$$

Here $N_c = 3$ is the number of colors, $N_c^2 = 9$ averages over the two initial quark colors, and the $\frac{1}{16}$ comes from the $(\frac{1}{4})^2$ factors at the two vertices (already present in the amplitude squared).

Step 4a: Use Hermiticity. Since the Gell-Mann matrices are Hermitian ($\lambda^{a\dagger} = \lambda^a$), we have:

$$(\lambda^b_{ji})^* = \lambda^b_{ij}$$

So the color sum becomes:

$$\sum_{i,j,k,l} \sum_{a,b} \lambda^a_{ji} \lambda^b_{ij} \; \lambda^a_{lk} \lambda^b_{kl} \tag{13}$$

Step 4b: Recognize the traces. Each pair of λ matrices summed over both indices is a trace:

$$\sum_{i,j} \lambda^a_{ji} \, \lambda^b_{ij} = \sum_j \Big(\sum_i \lambda^b_{ij} \, \lambda^a_{ji} \Big) = \sum_j \big(\lambda^b \lambda^a \big)_{jj} = \mathrm{Tr}(\lambda^b \lambda^a) \tag{14}$$

$$\sum_{k,l} \lambda^a_{lk} \, \lambda^b_{kl} = \mathrm{Tr}(\lambda^a \lambda^b) \tag{15}$$

The Key Identity: $\mathrm{Tr}(\lambda^a \lambda^b)$

This is the orthogonality relation for the Gell-Mann matrices (Griffiths 2nd ed., eq. 8.52; Thomson eq. 9.32). It says the eight λ^a matrices are "orthogonal" under the trace inner product, with normalization 2.

Quick check: Take $a = b = 3$. Then $\lambda^3 = \mathrm{diag}(1, -1, 0)$, so $(\lambda^3)^2 = \mathrm{diag}(1, 1, 0)$, and $\mathrm{Tr}(\lambda^3 \lambda^3) = 2$. ✓

For $a \neq b$, e.g. $a = 1, b = 3$: $\lambda^1 \lambda^3 = 0 - 10$
100
000, and Tr $= 0$. ✓

Step 4c: Assemble.

$$\begin{aligned} \sum_{\text{colors}} |C_{\text{amp}}|^2 &= \sum_{a,b} \mathrm{Tr}(\lambda^a \lambda^b) \, \mathrm{Tr}(\lambda^a \lambda^b) \\ &= \sum_{a,b} (2\delta^{ab})(2\delta^{ab}) = 4 \sum_{a,b} \delta^{ab} \delta^{ab} \\ &= 4 \sum_{a=1}^{8} 1 = 32 \end{aligned} \tag{16}$$

Step 4d: Include prefactors.

$$\langle |C|^2 \rangle = \frac{1}{9} \cdot \frac{1}{16} \cdot 32 = \frac{32}{144} = \frac{2}{9} \tag{17}$$

Color-Averaged Color Factor

$$\boxed{\langle |C|^2 \rangle = \frac{2}{9}} \tag{18}$$

This result is universal for all four t-channel single-gluon-exchange processes (qq, $q\bar{q}$, $\bar{q}q$, $\bar{q}\bar{q}$), as confirmed by Thomson (eq. 10.29).

6.2 Option B: Specific Color Choices

Instead of averaging, we can pick definite colors and compute the color factor at the *amplitude* level:

$$C = \frac{1}{4} \sum_{a=1}^{8} \lambda^a_{ji} \, \lambda^a_{lk}$$

(The $\frac{1}{4}$ comes from the two vertex factors each carrying $\frac{1}{2}$.)

Example: Red quark + Green quark → Red + Green (elastic)

With $i = 1$ (red), $j = 1$ (red), $k = 2$ (green), $l = 2$ (green):

$$C = \frac{1}{4} \sum_{a} \lambda^a_{11} \, \lambda^a_{22}$$

Scanning the Gell-Mann matrices, only $\lambda^3 = \text{diag}(1, -1, 0)$ and $\lambda^8 = \frac{1}{\sqrt{3}}\text{diag}(1, 1, -2)$ have non-zero entries in both the 11 and 22 positions:

$$\begin{aligned} C &= \frac{1}{4}\left[\lambda^3_{11}\lambda^3_{22} + \lambda^8_{11}\lambda^8_{22}\right] = \frac{1}{4}\left[(1)(-1) + \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}}\right] \\ &= \frac{1}{4}\left[-1 + \frac{1}{3}\right] = \frac{1}{4}\left(-\frac{2}{3}\right) = -\frac{1}{6} \end{aligned} \tag{19}$$

This matches Thomson eq. 10.33: $C(rg \to rg) = -\frac{1}{6}$.

Example: Red quark + Red quark → Red + Red (elastic)

With $i = j = k = l = 1$:

$$C = \frac{1}{4} \sum_{a} (\lambda^a_{11})^2 = \frac{1}{4}\left[(1)^2 + (1/\sqrt{3})^2\right] = \frac{1}{4} \cdot \frac{4}{3} = \frac{1}{3}$$

This matches Thomson eq. 10.32: $C(rr \to rr) = \frac{1}{3}$.

Physical Meaning of the Color Factor Sign

The sign of C tells you about the nature of the force in that color channel:

- $C = +\frac{1}{3}$ for $rr \to rr$: repulsive (like charges repel, as in QED).

- $C = -\frac{1}{6}$ for $rg \to rg$: attractive! (This is the "triplet" channel — the antisymmetric color combination $\frac{1}{\sqrt{2}}(rg - gr)$ that ultimately builds baryons.)
- For color-exchange scattering $rg \to gr$: $C = +\frac{1}{2}$ (Thomson eq. 10.34).

When you square these and average over all 9 initial colors, summing over all 9 final colors, you get $\frac{2}{9}$ in every case.

7 Step 5: Putting It All Together

Now we assemble the pieces. From eq. (**??**):

$$\langle|\mathcal{M}|^2\rangle_{\text{QCD}} = \frac{g_s^4}{16} \times \langle|C|^2\rangle \times \frac{1}{4} \times \frac{8(s^2+u^2)}{t^2} \tag{20}$$

Wait — let's be very careful about what's already included in $\langle|C|^2\rangle = 2/9$.

Careful Factor Accounting

The amplitude squared, summed over all spins AND all colors, is:

$$\sum_{\text{all}} |\mathcal{M}|^2 = \frac{g_s^4}{16} \times \underbrace{\sum_{a,b,i,j,k,l} \lambda^a_{ji}\lambda^b_{ij}\lambda^a_{lk}\lambda^b_{kl}}_{=32} \times \frac{8(s^2+u^2)}{t^2}$$

To get the *averaged* quantity, we divide by:

- $(2s_1+1)(2s_2+1) = 2 \times 2 = 4$ for spin averaging,
- $N_c \times N_c = 3 \times 3 = 9$ for color averaging.

So:

$$\begin{aligned}\langle|\mathcal{M}|^2\rangle &= \frac{1}{4} \cdot \frac{1}{9} \cdot \frac{g_s^4}{16} \cdot 32 \cdot \frac{8(s^2+u^2)}{t^2} \\ &= \frac{g_s^4}{36} \cdot 2 \cdot \frac{8(s^2+u^2)}{t^2} \\ &= \frac{g_s^4}{36} \cdot \frac{16(s^2+u^2)}{t^2}\end{aligned} \tag{21}$$

Computing: $\frac{16}{36} = \frac{4}{9}$.

Final Result: $\langle|\mathcal{M}|^2\rangle$ for $q\,q' \to q\,q'$ via Gluon Exchange

$$\boxed{\langle|\mathcal{M}|^2\rangle_{qq'\to qq'} = \frac{4}{9}\, g_s^4\, \frac{s^2+u^2}{t^2}} \tag{22}$$

Equivalently, in terms of $\alpha_s = g_s^2/(4\pi)$:

$$\langle|\mathcal{M}|^2\rangle = \frac{4}{9}\,(4\pi\alpha_s)^2\,\frac{s^2+u^2}{t^2} \tag{23}$$

Cross-check with QED:

$$\frac{\langle|\mathcal{M}|^2\rangle_{\text{QCD}}}{\langle|\mathcal{M}|^2\rangle_{\text{QED}}} = \frac{\frac{4}{9}\,g_s^4}{2\,g_e^4} = \frac{2}{9}\,\frac{g_s^4}{g_e^4}$$

The $\frac{2}{9}$ is precisely the color-averaged color factor $\langle|C|^2\rangle$, confirming that the only changes from QED to QCD are the coupling constant replacement and the color factor.

8 Summary: The Recipe for QED → QCD Conversion

What You Should Take Away

1. **Same Dirac algebra.** Any QED t-channel spin-$\frac{1}{2}$ scattering amplitude involving a massless spin-1 mediator has its spinor-trace structure captured entirely by s^2+u^2 in the massless limit. Switching from photon to gluon doesn't change this.

2. **Coupling replacement.** At each QCD vertex, $g_e \to \frac{g_s}{2}\lambda^a$. The $\frac{1}{2}$ is Griffiths' convention (absorbing it into $T^a = \lambda^a/2$ gives the same physics).

3. **Color factor computation.**

   - The color matrices λ^a_{ji} are just numbers in 3×3 color space — they commute freely with the 4×4 Dirac matrices.
   - Squaring and summing over colors produces traces $\text{Tr}(\lambda^a\lambda^b) = 2\delta^{ab}$.
   - The color-averaged result is $\langle|C|^2\rangle = \frac{2}{9}$ for any single-gluon-exchange process.

4. **Different flavors ⇒ t-channel only.** If the quarks had the same flavor, we'd need a second (u-channel) diagram, antisymmetrized via Feynman Rule 8. The problem avoids this by specifying different flavors.

5. **The parton model context.** This $\langle|\mathcal{M}|^2\rangle$ is the "hard scattering" piece of a hadron–hadron cross section. In a full calculation, you'd convolve it with parton distribution functions and include fragmentation — but the problem explicitly says to ignore those.